

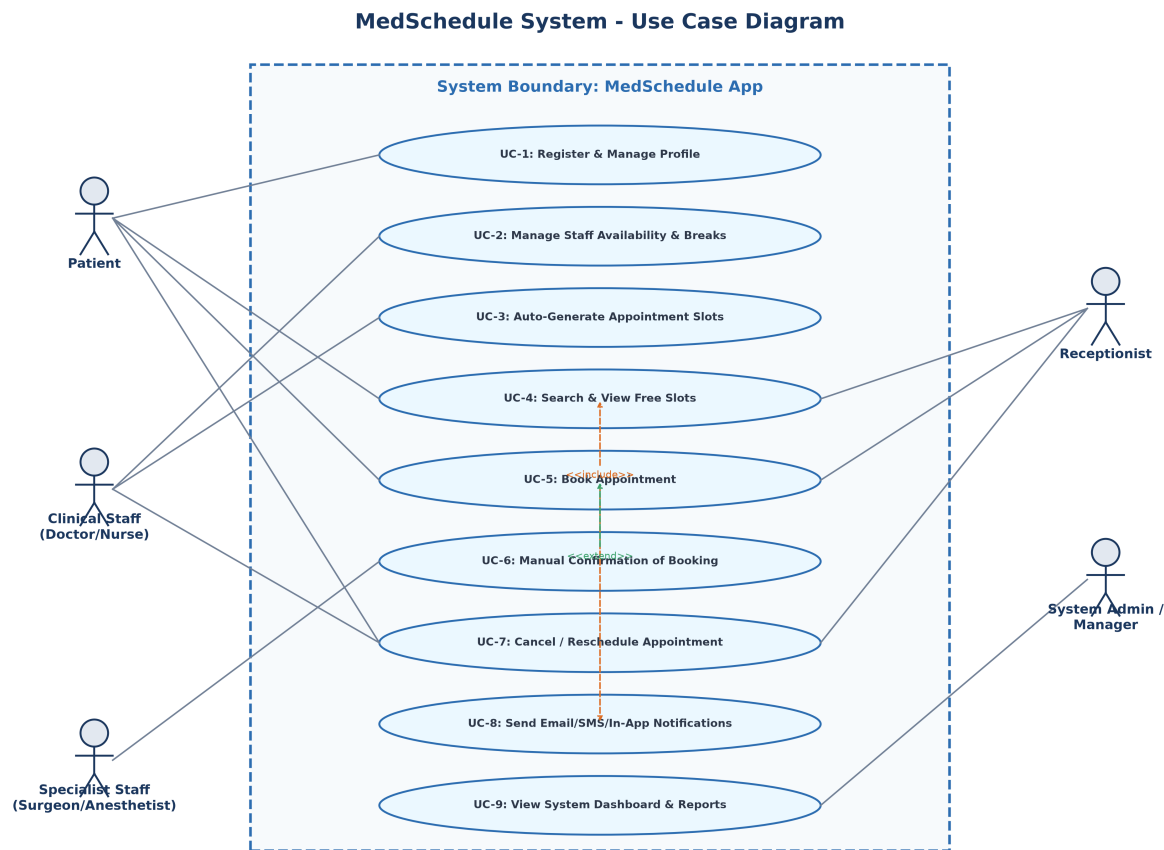
Deliverable 2: Use Case Diagram & Detailed Description

Project: MedSchedule Healthcare Scheduling System | Date: August 2026

1. System Use Case Overview & Architecture

The Use Case model defines the functional boundaries of the MedSchedule platform. It captures interactions between external actors (Patients, Clinical Staff, Specialists, Administrative Users) and core system services including slot generation, appointment booking, approval workflows, and multi-channel notifications.

2. System Use Case Diagram



3. Actor Glossary & System Roles

Actor Name	Actor Category	Description & System Interaction
Patient	Primary Human Actor	Searches available slots, books appointments, manages personal medical profile.
Clinical Staff	Primary Human Actor	Doctor, Nurse, Pharmacist, Lab Scientist, etc. Sets weekly schedules and conducts appointments.
Specialist Staff	Primary Human Actor	Surgeon, Anesthetist. Requires manual review of pending patient bookings.
Receptionist	Primary Human Actor	Administrative staff booking appointments on behalf of walk-in patients.
Admin / Manager	Primary Human Actor	Oversees system operations, views performance analytics, calls staff meetings.
Notification Engine	Secondary System Actor	Automated service executing SMS/Email dispatches based on triggers.

4. Detailed Use Case Specifications

Use Case ID: UC-01	Use Case Name: Book Appointment
Primary Actor	Patient / Receptionist
Supporting Actors	Clinical Staff (Doctor), Notification Engine
Preconditions	User is authenticated. Target staff member has active available appointment slots.
Postconditions	Appointment record is created; slot status updated to 'booked'; notifications dispatched.
Main Success Scenario	1. User selects clinical staff role, specific provider, date, and time slot. 2. System verifies slot status is 'available'. 3. User enters appointment reason and submits. 4. System creates Appointment object and links it to AppointmentSlot. 5. If provider is standard staff, status = 'confirmed'. If Surgeon/Anesthetist, status = 'pending'. 6. System invokes send_notification() for SMS/Email alert. 7. System redirects user to dashboard with success message.
Extensions / Exceptions	3a. Slot already booked by another user (race condition): System alerts user to choose another slot. 5a. Manual confirmation required: System flags appointment as pending approval by specialist.